

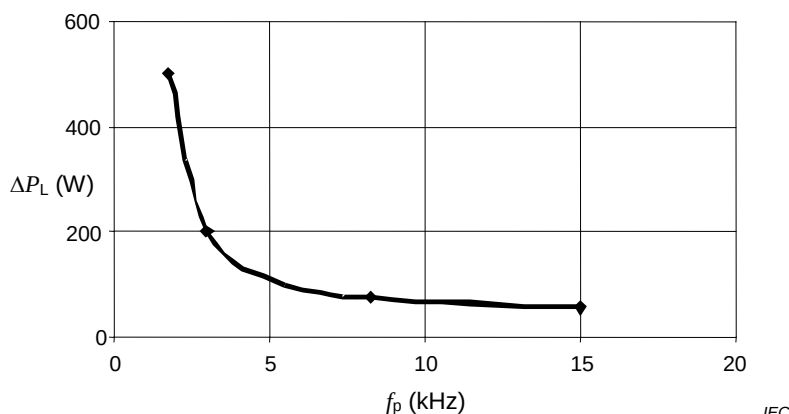


Figure 6 – Additional losses ΔP_L of an electrical machine (same electrical machine as Figure 5) due to converter supply, as a function of pulse frequency f_p , at 50 Hz rotational frequency

5.4 Use of filters to reduce additional electrical machine losses due to converter supply

Filters may be used at the output from a converter to reduce the amplitude and du/dt of the high-frequency switching voltage without excessively affecting the low-frequency resultant voltage appearing at the electrical machine terminals. The total effects will depend on the application and dimensioning of the electrical machine and the filter. The voltage drop across the filter will reduce the voltage at the electrical machine terminals, and this should be taken into account in order to avoid an increase in the fundamental current loss in the electrical machine. Also, there will be some losses in the filter, but these will generally be lower than the reduction of additional electrical machine losses due to converter supply, and so the overall efficiency of the PDS will improve.

In addition to reducing the additional electrical machine losses due to converter supply, such filters may also have a beneficial effect in reduced voltage stress on the electrical machine windings, decreased torque ripple, and improved EMC performance (see 9.2). However, there will be a slowing of the dynamic response of the PDS, and there may be other limitations due to the voltage drop across the filter.

5.5 Temperature influence on life expectancy

The sum of the fundamental and additional losses due to the load condition and the voltage waveform results in a temperature rise of the electrical machine windings. The temperature rise will also be affected by a change in cooling at the operating point within the specified speed range.

There are several ways to take this effect into account, for example:

- use of a separate cooling supply, such as IC0A6 or IC1A7 (see IEC 60034-6) for an air-cooled electrical machine;
- use of a higher thermal insulation class (see IEC 60034-1);
- full compensation for the intended operating ambient temperature (see IEC 60034-1);
- use of oversized electrical machine;
- optimisation of converter output waveform.

NOTE Increased temperatures may affect not only the winding insulation ageing but also the bearing lubrication, and hence the bearing lifetime.

The influence of variable load and speed on the winding temperature is characterized by the duty type as defined in IEC 60034-1. The most suitable duty types for converter-fed electrical

machines are S1 and S10. Duty type S1 considers the maximum permitted temperature, whereas S10 (for operation at varying load and speed) permits temperature rises which exceed the limit values of the thermal class for limited periods. Limit values of temperature rise are given in IEC 60034-1:2010, and Annex A of that standard gives a formula for the calculation of thermal life expectancy.

5.6 Determination of electrical machine efficiency

The recommended methods to determine the electrical machine efficiency on sinusoidal supply are given in IEC 60034-2-1 and IEC 60034-2-2, For voltage converter fed cage induction motors a new method to determine harmonic losses is described in IEC TS 60034-2-3.

If practical, if required to achieve a more accurate assessment of the overall losses (including the additional harmonic losses), they should be determined with the behaviour of the final PDS.

6 Acoustic noise, vibration and torsional oscillation

6.1 Acoustic noise

6.1.1 General

The converter and its function create three variables which directly affect emitted noise. They are:

- changes in rotational speed which may range from near zero speed to values in excess of the corner point speed. The components and factors that influence noise emissions are bearings and lubrication, ventilation and any other features that are affected by temperature changes;
- electrical machine power supply frequency and harmonic content which have a large effect on the magnetic noise excited in the stator core and, to a lesser extent, on the bearing noise;
- torsional and radial excitations of the stator core due to the interaction of waves of different frequencies of the magnetic field in the electrical machine air gap.

6.1.2 Changes in noise emission due to changes in speed

6.1.2.1 Sleeve (or plain) bearings

There will be no significant change in the noise level emitted by plain bearings.

6.1.2.2 Rolling element bearings

The fundamental frequencies of potential noise emission from a rolling element bearing will vary directly with the rotational speed. If the bearing is "quiet" at the corner point speed, it is unlikely for the noise level to change significantly when the speed is reduced. However, when the speed is increased above the corner point speed there is the possibility that the noise level could increase dramatically due to harmonics of the fundamental frequencies growing due to skidding of the rolling elements. The susceptibility to this phenomenon has been shown to increase rapidly at speed factors (bearing diameter in mm $\times$ rotational speed in r/min) greater than 180 000. Experience has shown that the noise level increase can be countered by increasing the lubricant supply to the bearing by regreasing at very short intervals or by utilising oil bath or oil mist lubrication.

When operating at the highest speeds in the electrical machine's range, the bearing temperature will be higher than running at lower speeds. It is important therefore to ensure that adequate nominal clearance and/or a spring loaded arrangement is embodied in the design.

Grease lubricated bearings will perform perfectly satisfactorily at low operating speeds.

6.1.2.3 Ventilation noise

For a shaft-mounted fan, the noise generated will vary approximately as the characteristic shown in Figure 7 (for a fan peripheral velocity up to 50 m/s). The fan noise will decrease by about 15 dB for a 50 % reduction in speed and increase by about 10 dB for a speed increase of 50 %. If the drive is unidirectional, very effective noise reduction can be achieved by utilising a fan on the electrical machine with curved unidirectional blades.

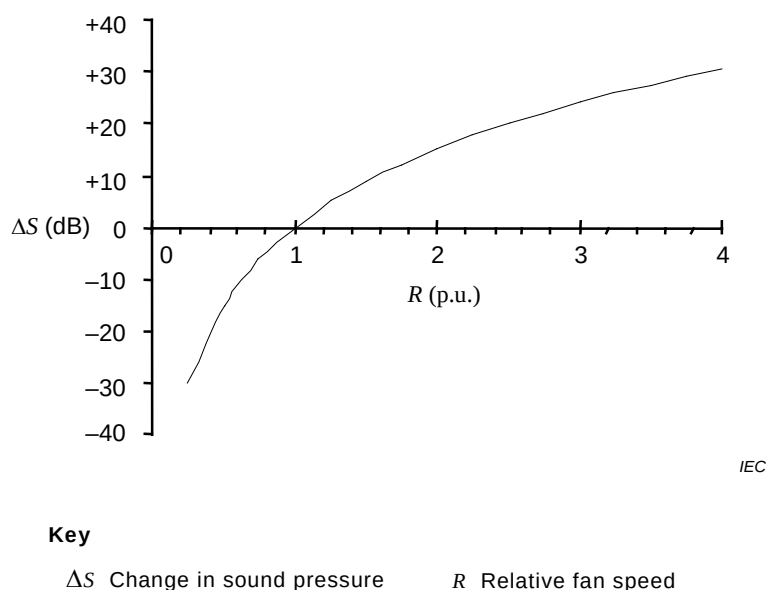


Figure 7 – Relative fan noise as a function of fan speed

6.1.3 Magnetically excited noise

Magnetic noise is essentially caused by waves of tensile stress acting in the radial direction on the stator bore. These so-called Maxwell forces are excited by the interaction of the various magnetic fields in the air-gap. The tensile stress is characterized by its amplitude, frequency and mode. As the amplitudes are small, the tensile stress results in disturbing tones only when frequency and mode of a specific wave coincides with the frequency and mode of a natural frequency of the stator core.

In the case of sinusoidal supply voltages, the magnetic noise is caused by the spatial harmonics of the air-gap field. The aim of a professional design is to avoid resonances at the rated operating conditions of the electrical machine. But, because of the large variety of contributing spatial harmonic fields, audible magnetic noise is unavoidable at specific speeds, when the electrical machine is operated at constant flux over a wide speed setting range, even when the supply voltage is sinusoidal. Skipping of a small frequency band is frequently used to avoid a too high noise emission at the associated speed. This means this effect is not associated with the converter supply and would also occur in case of variable-frequency sinusoidal supply voltages.

The statements given above are valid also when an electrical machine is supplied from a converter. But, in this case, the magnetic fields produced by the time harmonics are superimposed. With respect to considerable magnetic noise, it is sufficient to consider the interaction of the fundamental air-gap fields (number of pole pairs p) of the operating frequency and the different harmonics. Therefore, the additionally generated waves of tensile stress are of the modes $r = 0$ and $r = 2p$. The natural frequencies of these modes depend on the size and the design of the electrical machine

The objective of *PDS* designers is to create optimum noise solutions, but it should be recognized that such solutions are not the responsibility of either the converter designer or the electrical machine designer alone and that in many cases design co-operation is essential.

Experience has shown that with pulse frequencies less than 3 kHz, the harmonic frequencies can be close to the natural frequencies of the electrical machine core and structure on medium and large electrical machines and consequently with wide speed range applications, resonance points are nearly unavoidable at some point in the speed range (see Figure A.2). The resonance frequencies for the modes $r = 0$ and $r = 2p$ (see Figure 8 for illustrations of modes $r = 0, 2$ and 4) are less than 2,5 kHz for 2-pole and 4-pole electrical machines with shaft height greater than 315 mm. By contrast, the trend to increase the converter pulse frequency to 4 kHz or 5 kHz or even higher will result in possible resonance occurring on progressively smaller electrical machines.

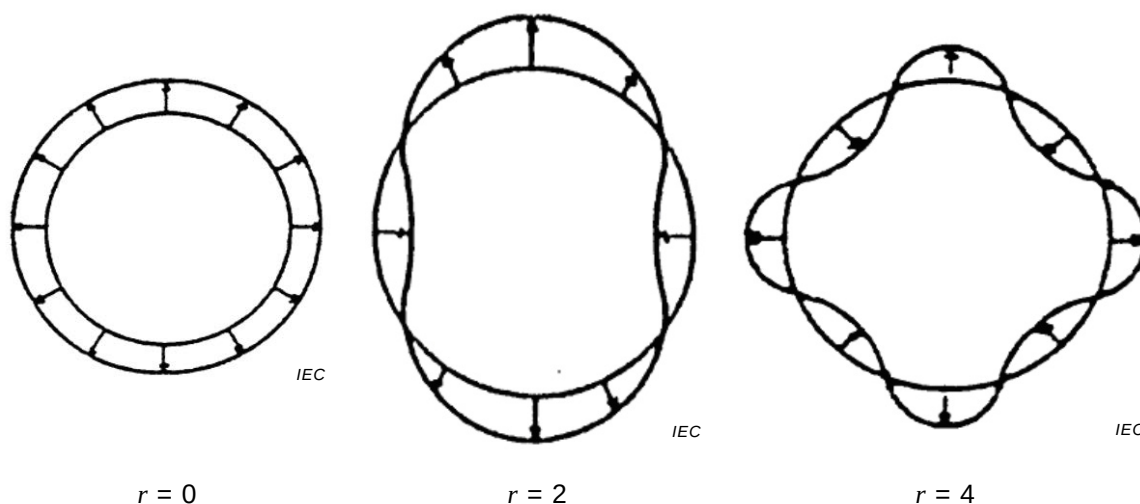


Figure 8 – Vibration modes of the stator core

The increment of noise of electrical machines supplied from PWM controlled converters compared with the same electrical machine supplied from a sinusoidal supply is relatively small (a few dB(A) only) when the switching frequency is above about 3 kHz. For lower switching frequencies, the noise increase may be tremendous (up to 15 dB(A) by experience). Some advanced PWM or hysteresis controlled converters no longer use fixed carrier frequencies and therefore produce a widely spread spectrum of non-fundamental frequencies. Thus, the typical noise increase and the subjective audible noise can be drastically reduced.

It may be necessary to create "skip bands" in the operating speed range in order to avoid resonance conditions.

For an indication of the noise increase when operated on a converter see IEC 60034-9:2007, Clause 7.

6.1.4 Sound power level determination and limits

6.1.4.1 Methods of measurement

Sound power levels should be determined in accordance with IEC 60034-9 (see 6.1.4.2).

6.1.4.2 Test conditions

If practical the electrical machine should be rigidly mounted and, tests should be made with the electrical machine supplied from a converter with the output characteristics that will be used in the application.